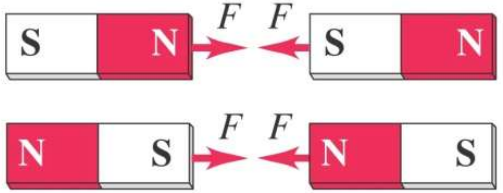
Magnetic field and forces

Magnetism

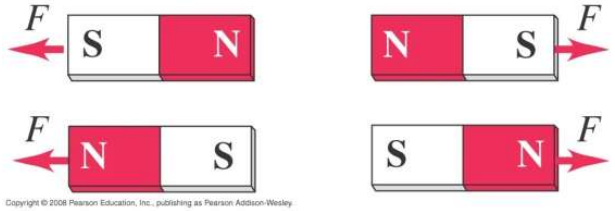
- Magnetic phenomena were first observed more than 2500 years ago in Turkey.

- If a bar-shaped magnet is free to rotate, one end points north and is called the north pole (N pole).

(a) Opposite poles attract.



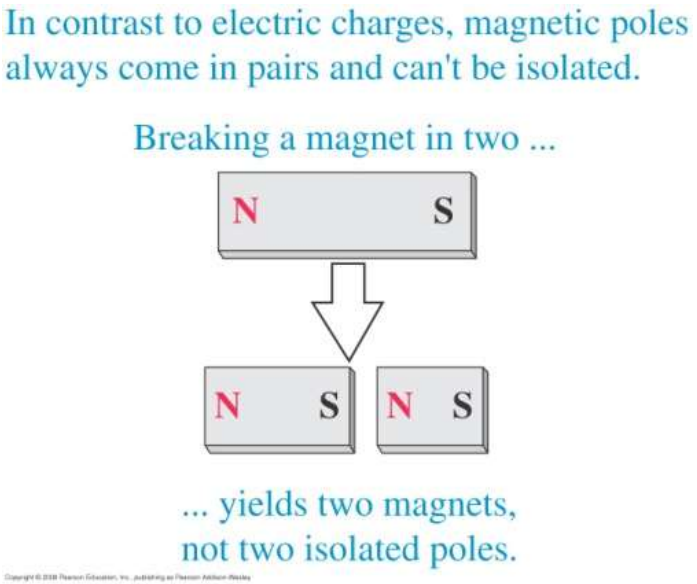
(b) Like poles repel.



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Magnetic poles always come in pairs

- If a bar magnet is broken into two parts, each broken end becomes a pole.

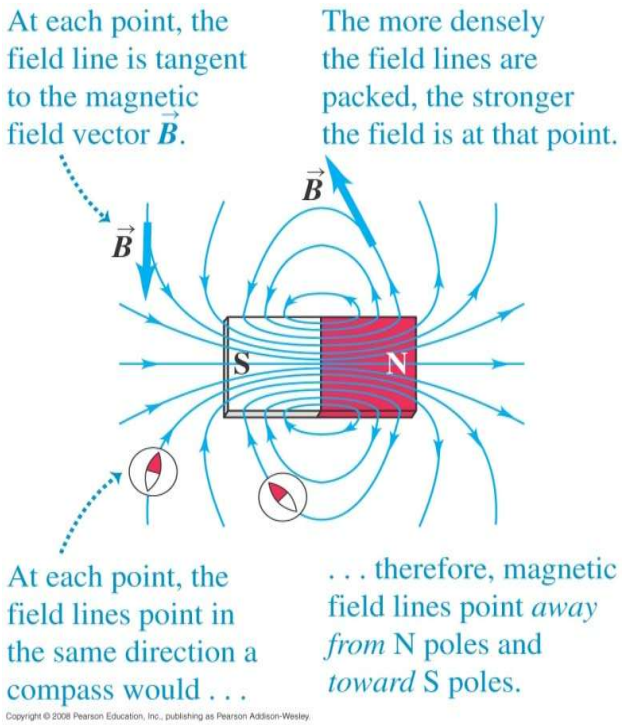


- There is no evidence of a single isolated magnetic pole or magnetic monopole.

Magnetic field

- A magnetic field is a region where a magnetic pole or a moving electric charge experiences a magnetic force.

- The magnetic field **B** at a given point is tangent to the field line at that point.



Magnetic forces on moving charges

- From experiments it has been found that the magnetic force on a moving charge is $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$.

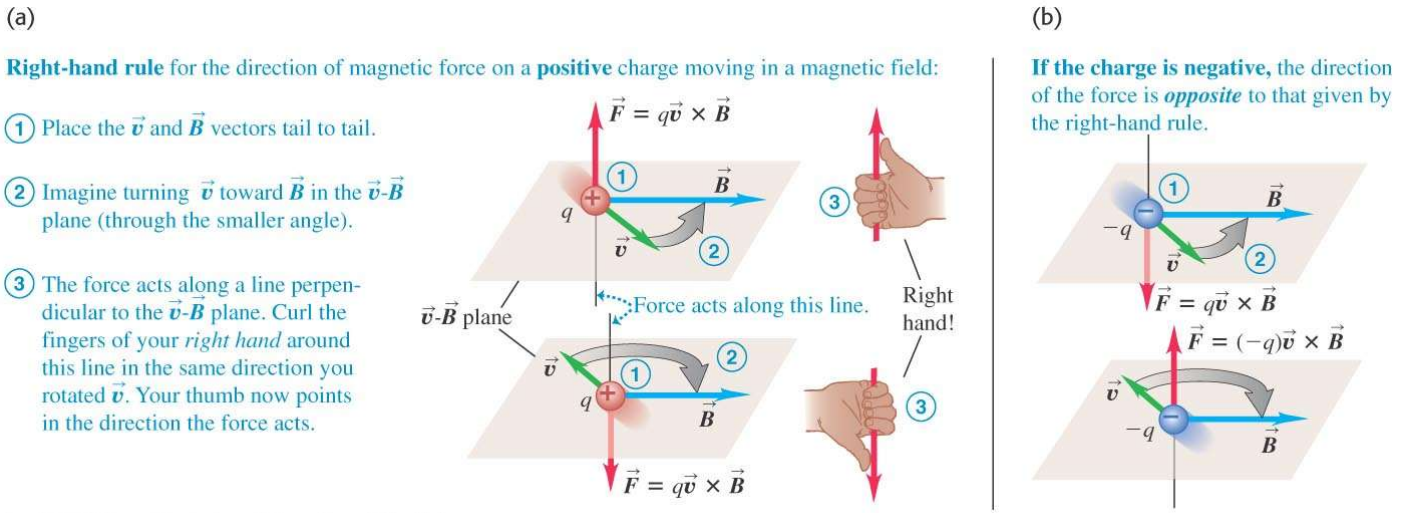
- The magnitude of **F** is $q|\mathbf{v}||\mathbf{B}|\sin(\phi)$ where ϕ is the angle between **v** and **B**.

- The direction of **F** is perpendicular to both **v** and **B**.

- The SI unit of B is N A^{-1} m^{-1} or tesla (T).

Right-hand grip rule to find the direction of F

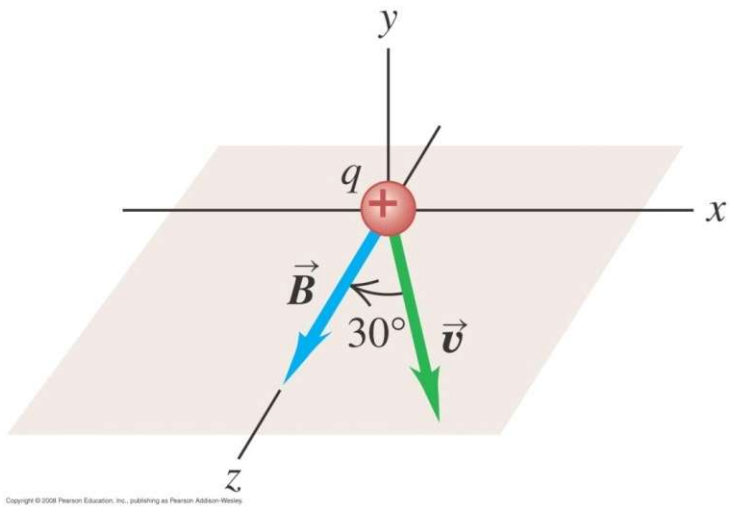
- Grip the velocity vector and move towards the magnetic field vector.
- The thumb gives the direction of the magnetic force on a positive charge.
- If the charge is negative the force is opposite to the one given by the thumb.



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Example 1

A beam of protons moves at 3.0 x 10^5 m/s through a uniform magnetic field with a magnitude 2.0 T. The magnetic field is directed along the positive *z*-axis as shown in the below figure. The velocity of each proton lies in the *x*-*z* plane at an angle of 30° to the positive *z*-axis. Find the force on a proton. Charge of a proton is 1.60 x 10^{-19} C.



Charge particle moving perpendicular to a uniform magnetic field

- The magnetic force is always perpendicular to the velocity and the particle moves with constant speed in a circle.

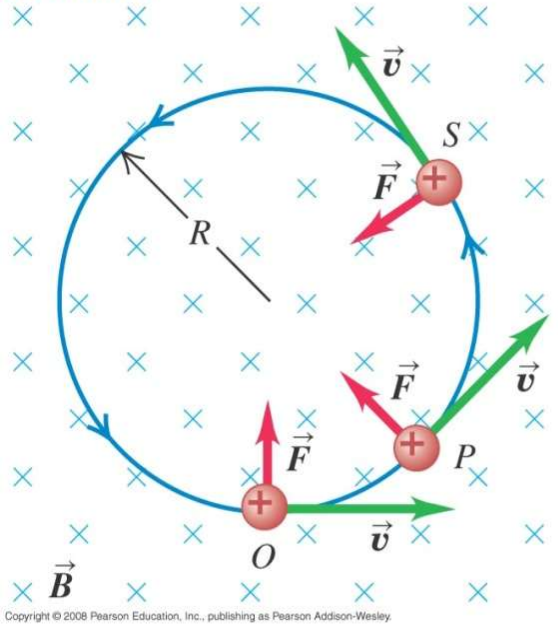
Animation

$$F = |q| vB = m\frac{v^2}{R}$$

and $$R = \frac{mv}{|q|B}$$

(a) The orbit of a charged particle in a uniform magnetic field

A charge moving at right angles to a uniform $\vec{B}$ field moves in a circle at constant speed because $\vec{F}$ and $\vec{v}$ are always perpendicular to each other.



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Cyclotron frequency

- The angular speed is

$$\omega = \frac{v}{R} = \frac{|q|B}{m}$$

- The frequency known as cyclotron frequency is

$$f = \frac{\omega}{2\pi} = \frac{|q|B}{2\pi m}$$

- The cyclotron frequency f does not depend on R.

- The time period $T = \frac{1}{f} = \frac{2\pi}{\omega} = \frac{2\pi m}{|q|B}$

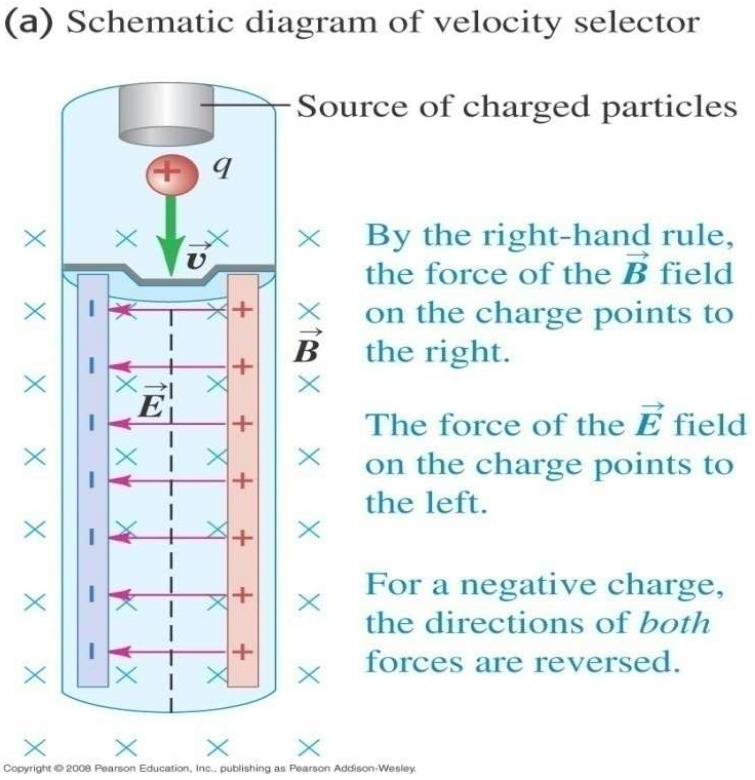
Velocity selector

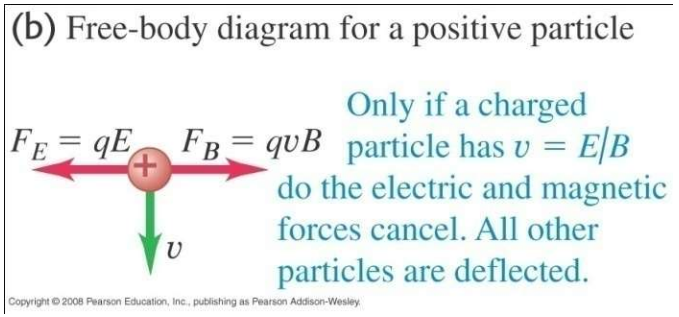
- Velocity selector is used to select positive and negative charges having a particular speed.

- E and B are perpendicular to each other and are such that the net force on the charged particle is zero. i.e.

$$qE = qvB$$

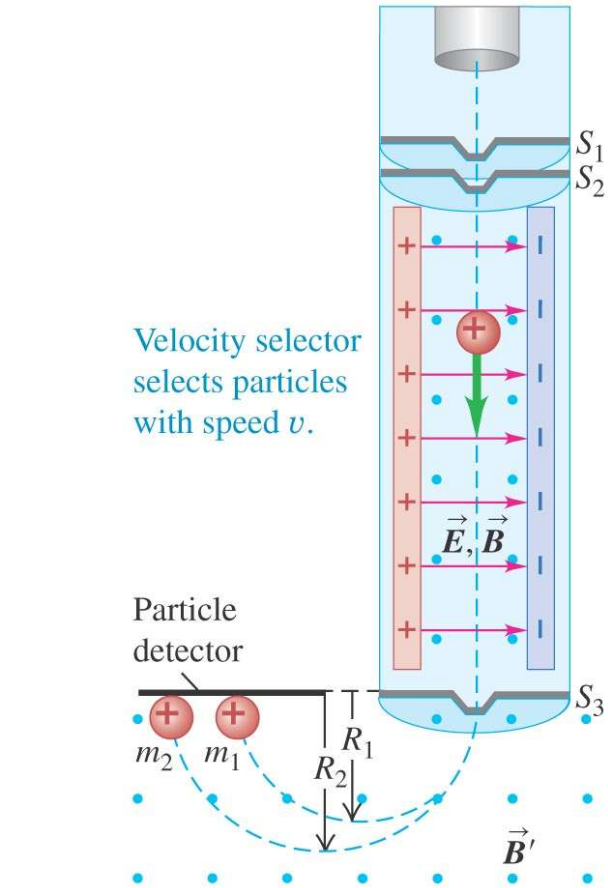
$$v = \frac{E}{B}$$





Mass spectrometer

- In a mass spectrometer, singly charged positive ions with velocity v = E/B pass through S_3.
- These ions then move into a region with a different magnetic field B' that is perpendicular to the figure.
- They move in circular arcs with radius R = mv/qB'.
- Isotopes of the same element (with different m) will move with different radii.



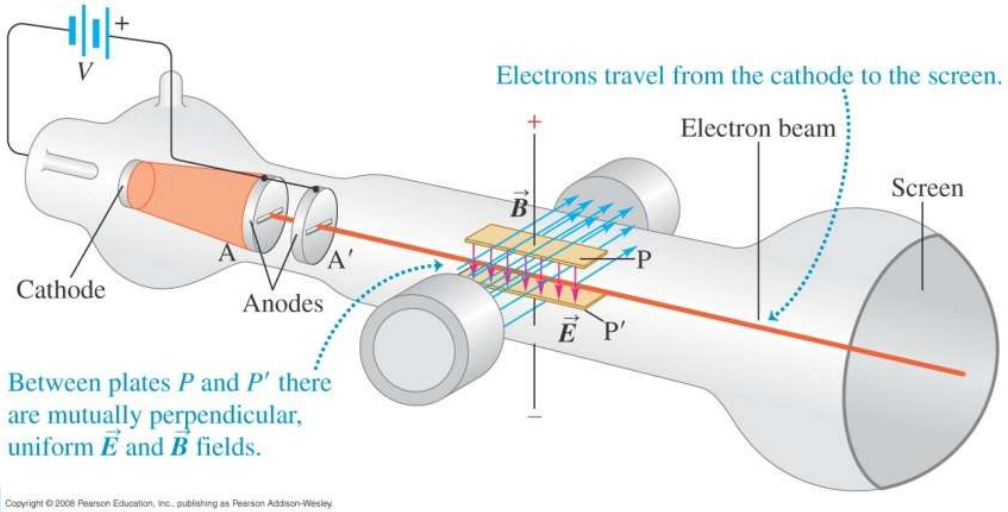
Magnetic field separates particles by mass; the greater a particle's mass, the larger is the radius of its path.

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Thomson's e/m experiment

- Electrons in a glass container are accelerated by a p.d. V between anodes A and A' and enter a magnetic field.

- The work done by the electric field on the electrons is eV, where e is the electronic charge. The electrons gain kinetic energy.



Thomson's e/m experiment - cont

- From conservation of energy,

$$\frac{1}{2}mv^2 = eV \quad \text{or} \quad v = \sqrt{\frac{2eV}{m}}$$

- Only electrons whose magnetic force equals its electric force will pass through and strike the screen, i.e.

$$eE = Bev$$

$$\text{or} \quad v = \frac{E}{B} = \sqrt{\frac{2eV}{m}}$$

$$\text{and} \quad \frac{e}{m} = \frac{E^2}{2VB^2}$$

- There is only a single value of e/m = 1.76×10^{11} C kg^{-1}.
- Since this value did not depend on the cathode material or any other thing in the experiment, the electrons in the beam is a common constituent of all matters.
- After some years Millikan succeeded in measuring the charge of the electron.
- Combining with Thomson's result and Millikan's experiments, the mass of the electron was determined to be about 9.11×10^{-31} kg.

Example 2

In a Thomson's e/m experiment, the accelerating potential is 150 V with a deflecting electric field of magnitude 6.0×10^6 N/C. Speed of light is 3×10^8 m/s. Mass of the electron = 9.11×10^{-31} kg.

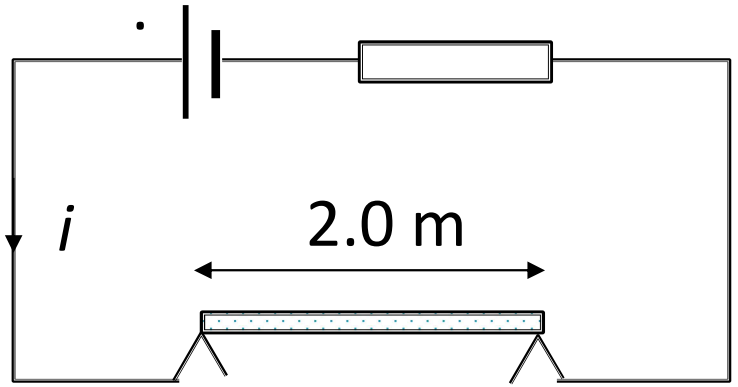
a) At what fraction of the speed of light do the electrons move?

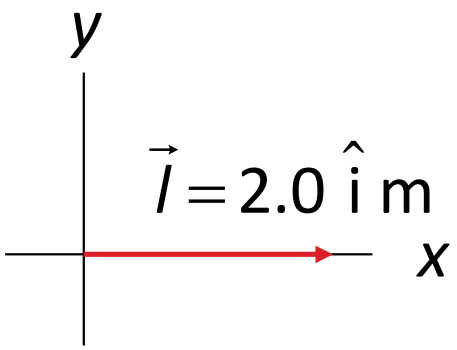
b) What should be the magnetic field?

c) With this magnetic field, what will happen to the electron beam if the accelerating potential is increased above 150 V?

Length as a vector

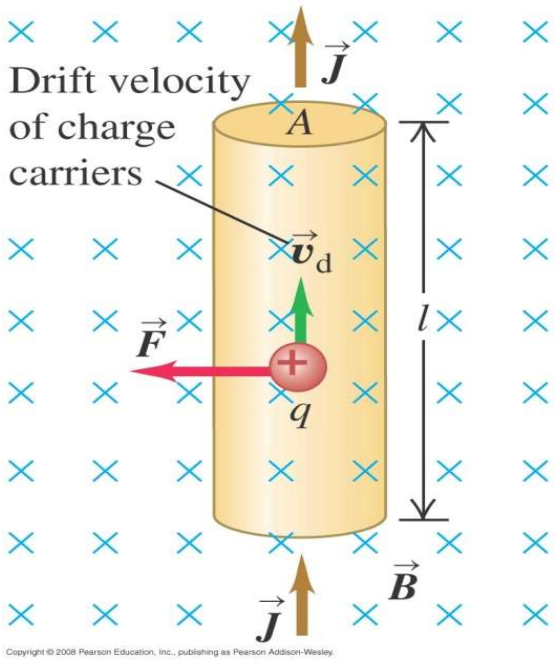
- The length of a conductor carrying a current can be regarded as a vector.
- The size of the length vector is the length of the conductor. Its direction is the direction of the current.
- In the circuit below, the direction of the length vector is in the direction of positive *x*, i.e. $\vec{l} = 2.0\ \hat{\mathrm{i}}\ \mathrm{m}$





Magnetic force on a current-carrying conductor

- Suppose positive charges flow with drift velocity $\vec{v}_d$ in a wire of length l and area A and perpendicular to an a uniform field $\vec{B}$.



- The total number of charges is nAl, where n = number of charges per unit volume.

- The total force on all the charges is

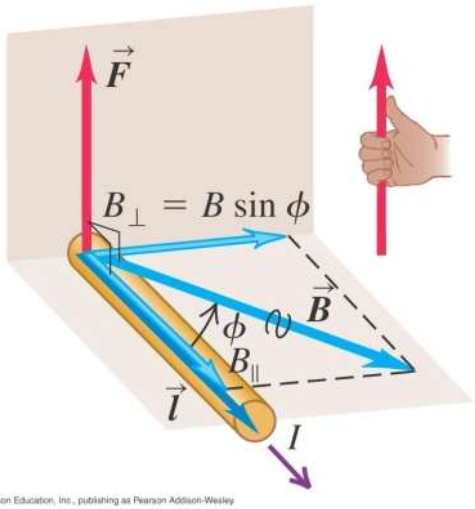
$$F = (nAl)(qv_dB)$$
$$= nqv_dAlB$$
$$= ilB \qquad \text{since } i = nqv_dA$$

Magnetic force on a current-carrying conductor

- If $\vec{B}$ makes an angle ϕ with the conductor, the force becomes

$$F = ilB\sin\phi$$

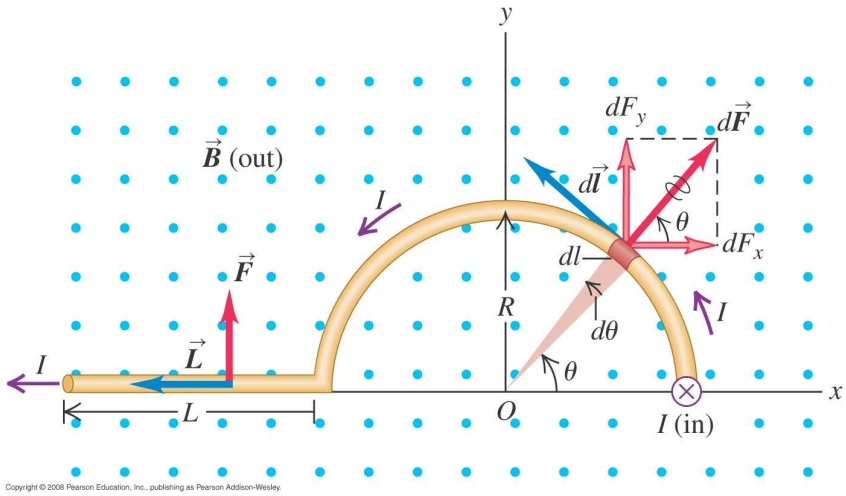
- In vector form $\vec{F} = i\vec{l}\times\vec{B}$



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- If the conductor is not straight, the force on each segment is

$$d\vec{F} = id\vec{l}\times\vec{B}$$



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- The total force is obtained by integrating over the whole length of the conductor.

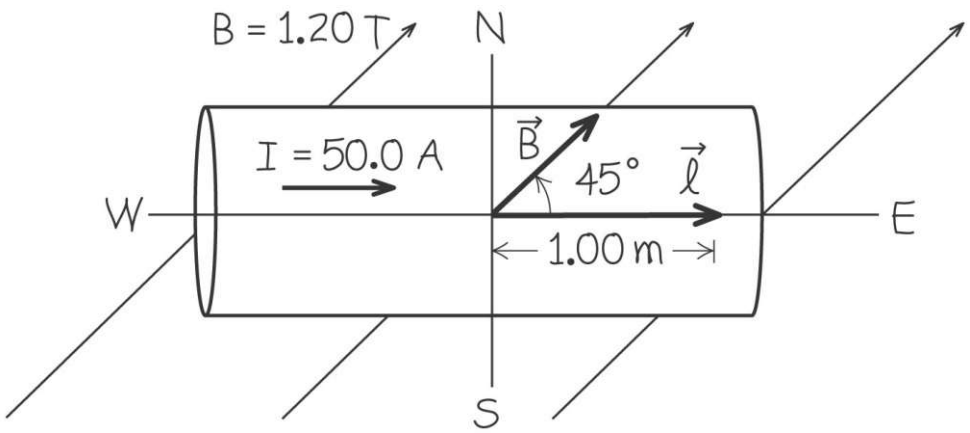
Example 3

A straight horizontal copper rod carries a current of 50.0 A from west to east in a region between the poles of a large electromagnet. In this region there is a horizontal magnetic field toward the north east (45° north of east) with magnitude 1.20 T.

a) Find the magnitude and direction of the force on a 1.00-m section of rod.

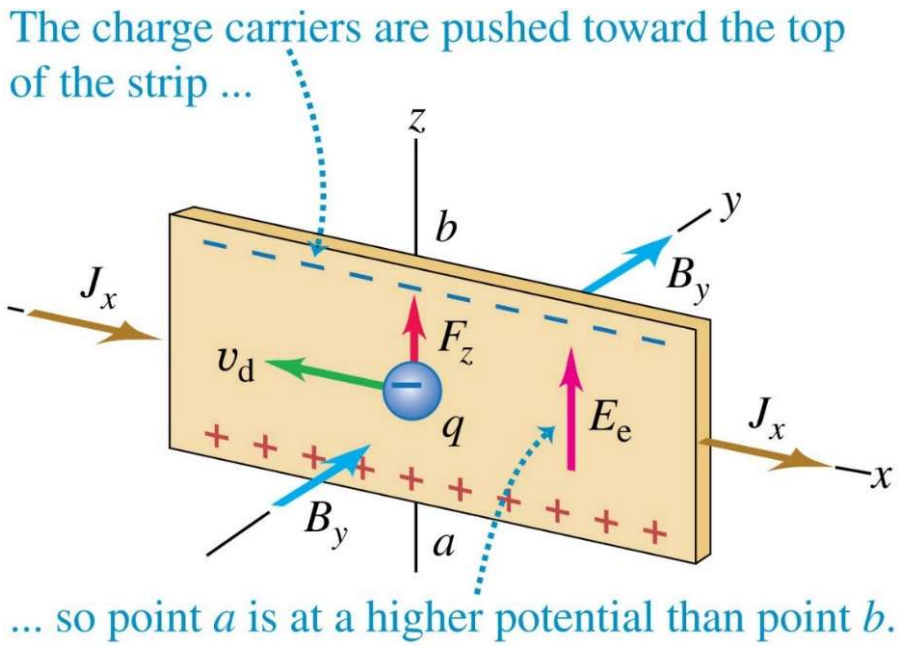
b) What should be done to the rod to maximize the magnitude of force?

c) What is the magnitude of the force in the above case in b)?



The Hall Effect: Negative Charge Carriers

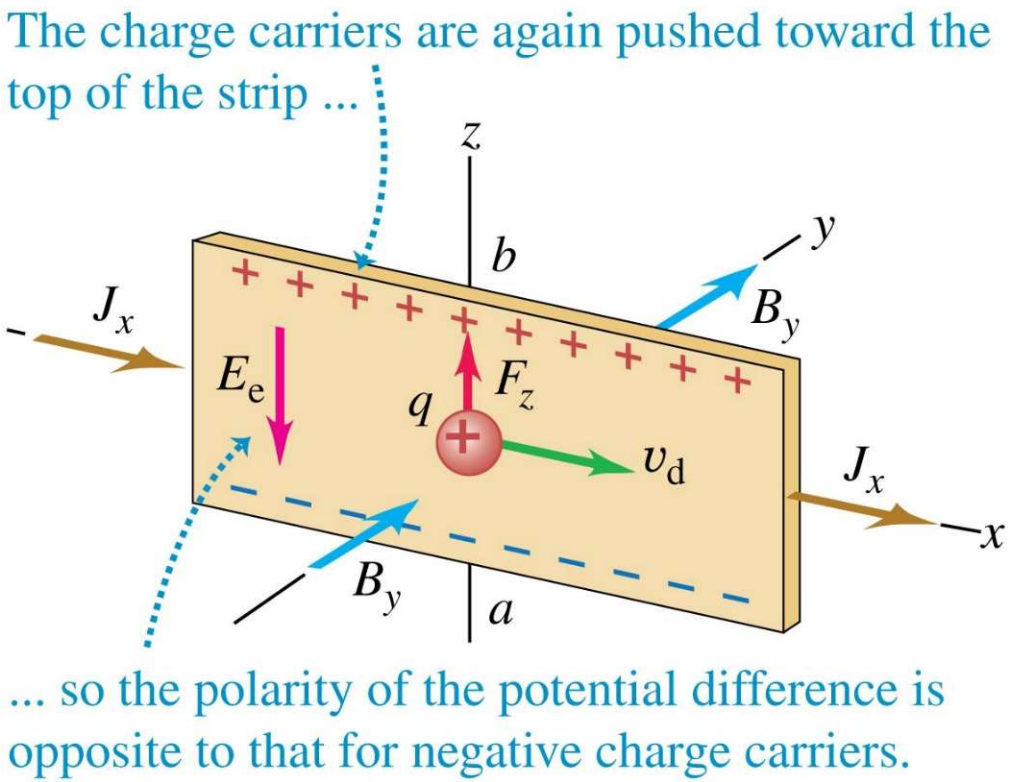
- When a current is placed in a magnetic field, the Hall emf reveals whether the charge carriers are negative or positive.



- Video Tutor Solution: Example 27.12

The Hall Effect: Positive Charge Carriers

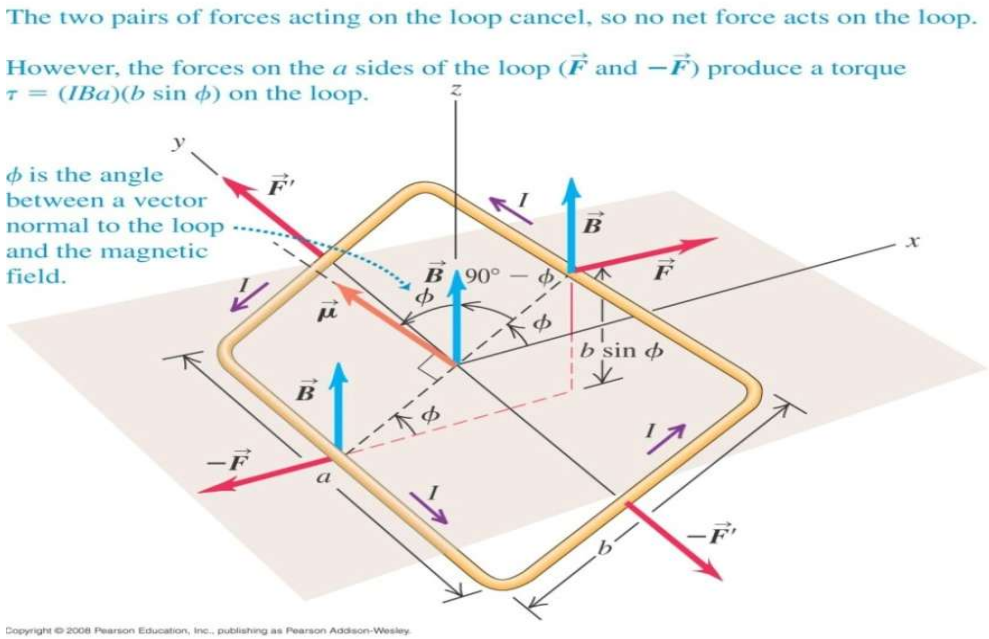
- When a current is placed in a magnetic field, the **Hall emf** reveals whether the charge carriers are negative or positive.



Force on a current loop

- The sides (length b) makes an angle of ($90^o - \phi$) with the direction of $\vec{B}$.

- $\vec{F}'$ and $-\vec{F}'$ have magnitude ibBcosϕ.

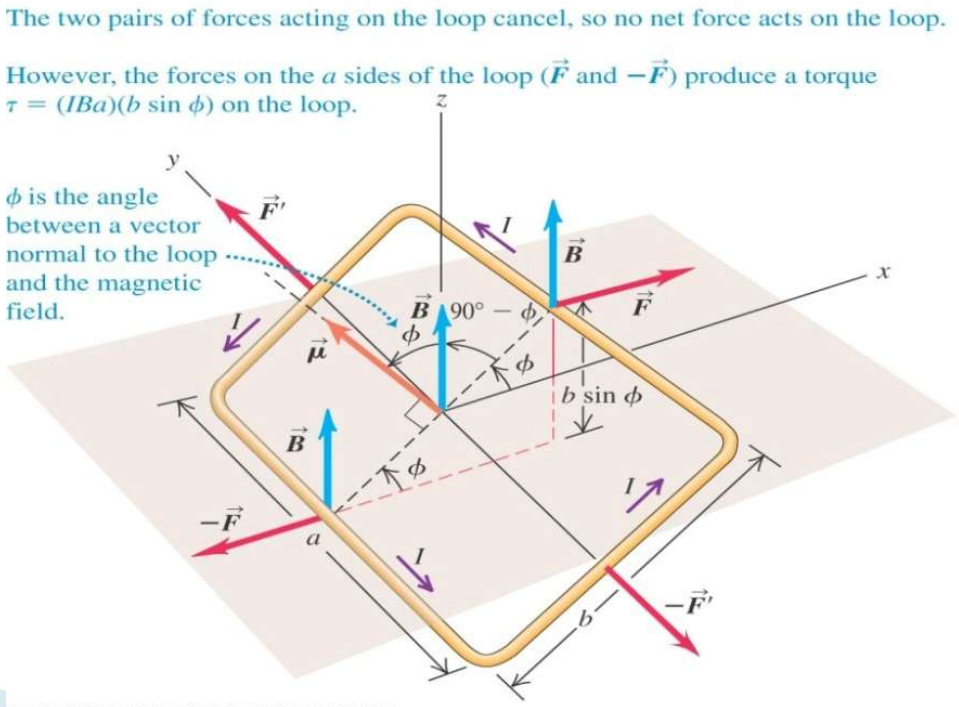


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Force on a current loop

- The force $\vec{F}$ on the right side of the loop (length a) is in the positive x-direction and has magnitude $F = iaB$.

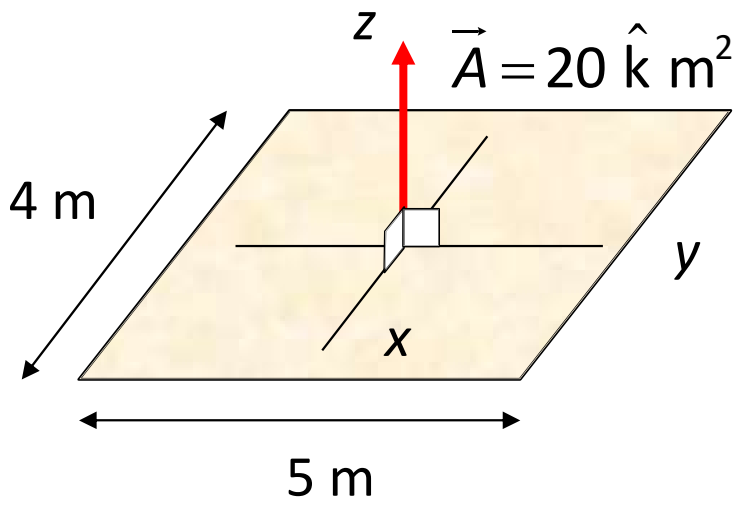
- A force $-\vec{F}$ with the same magnitude but opposite direction acts on the opposite side of the loop.



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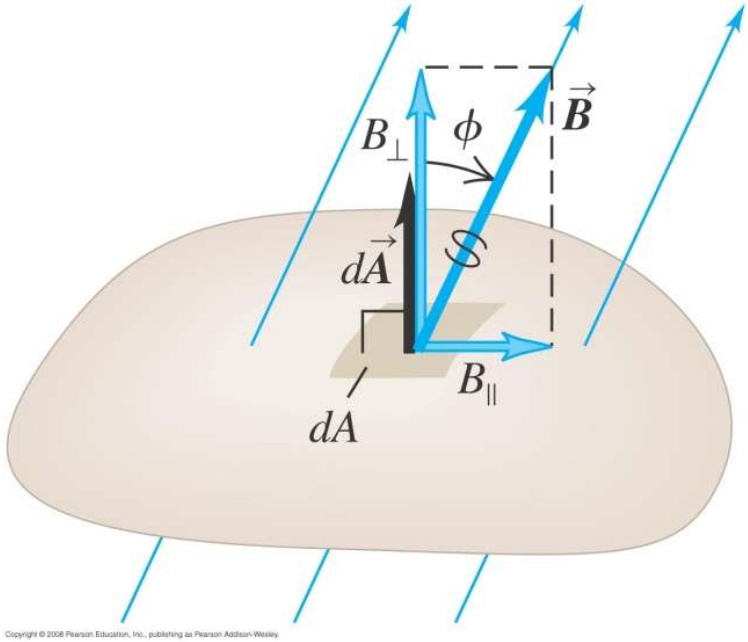
Area as a vector

- The area of a surface can be regarded as a vector.
- The size of the vector is the area occupied by $\vec{A}$ and its direction is the normal to the surface.
- In the example below, A = 20 m^2 and the direction of the area vector is in the direction of positive z.
- In terms of unit vector, $\vec{A} = 20\ \hat{k}\ \text{m}^2$



Magnetic flux

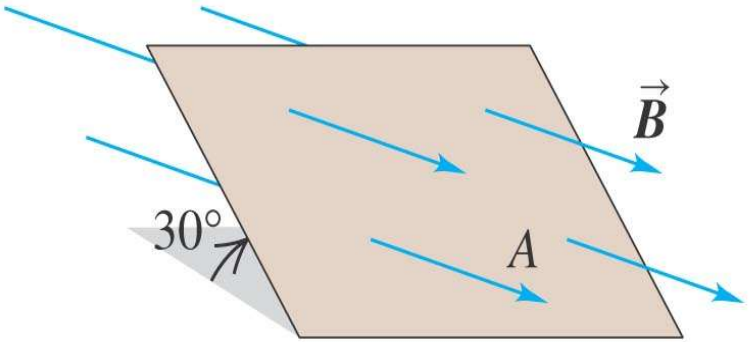
- The magnetic flux passing through an area dA is given by $d\Phi_B = \vec{B} \cdot d\vec{A}$
- The total magnetic flux through a surface A is given by $\Phi_B = \int \vec{B} \cdot d\vec{A}$
- The SI unit of magnetic flux is weber (Wb), where 1 Wb = 1 T m^2.
- Hence B is also known as flux density.



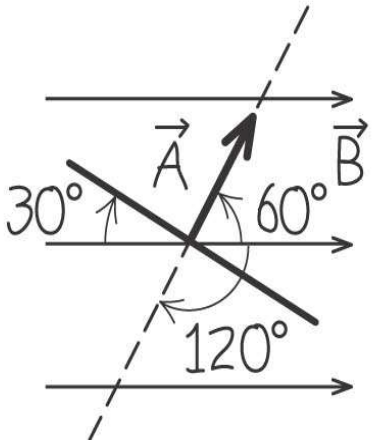
Example 5

The below figure shows a perspective view of a flat surface with area 3.0 cm^2 in a uniform magnetic field. If the magnetic flux through this area is 0.90 mWb, calculate the magnetic field and find the direction of the area vector.

(a) Perspective view



(b) Our sketch of the problem (edge-on view)



Gauss law of magnetism

- Magnetic field lines always form closed loops.
- Hence, every field line that enters the surface also exits from it.
- Since there is no magnetic monopole, the total magnetic flux through a closed surface is always zero. This is the Gauss's law of magnetism.
- Mathematically it is written as $\oint \vec{B} \cdot d\vec{A} = 0$

Torque on a current loop

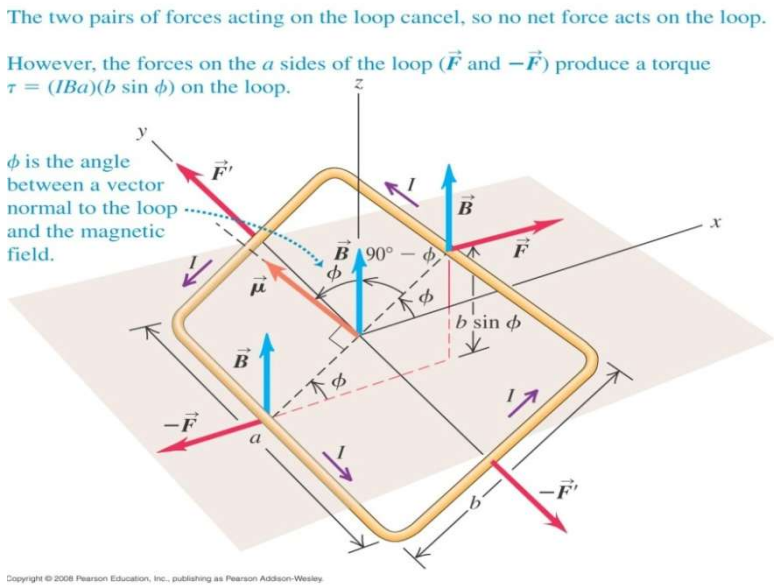
- $\vec{F}$ and $-\vec{F}$ lie along different lines and give rise to a torque which points in the positive y direction.
- By definition torque is, $\vec{\tau} = \vec{r} \times \vec{F}$

$$\left|\vec{\tau}\right| = 2F\left(\frac{b}{2}\right)\sin\phi = (iaB)(b\sin\phi)$$

$$= iAB\sin\phi \quad \text{where } A = \text{coil area}$$

$$= \mu B\sin\phi \quad \text{where } \mu = iA = \text{magnetic moment}$$

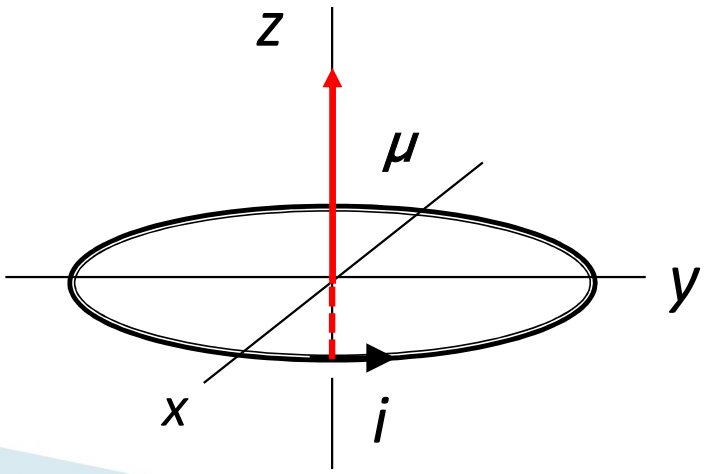
$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

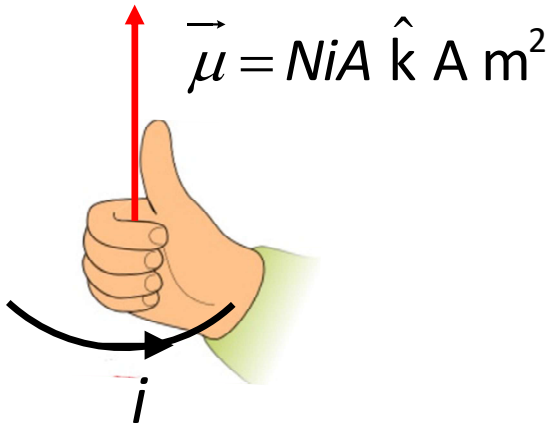


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Vector from a current loop

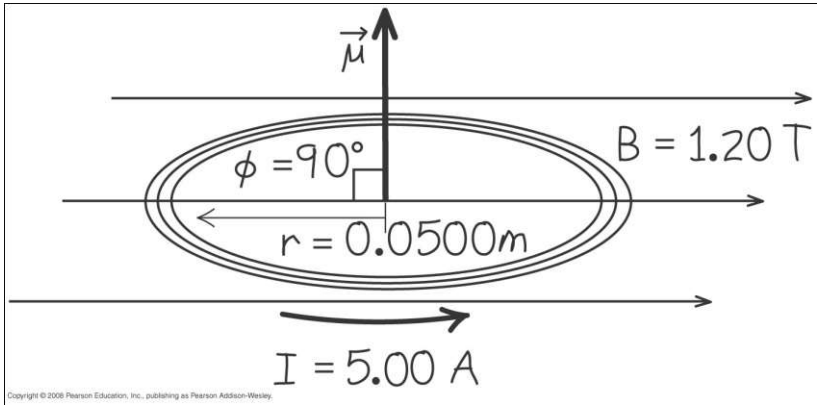
- The magnetic moment of a current loop can be regarded as a vector.
- The size of the magnetic moment vector is *NiA*, where *N* is number of turns in the loop, *i* is the current and *A* is the area of the loop.
- The direction of magnetic moment vector is given by the right hand grip rule, i.e. $\vec{\mu} = NiA\ \hat{\mathrm{k}}\ \mathrm{A\ m^2}$





Example 6

A circular coil 0.0500 m in radius, with 30 turns of wire, lies in a horizontal plane. It carries a current of 5.00 A in a counter-clockwise sense when viewed from above. The coil is in a uniform field directed towards the right, with magnitude 1.20 T. Find the magnitudes of the magnetic moment and the torque on the coil.



$$A = \pi(0.0500)^2 = 7.85\times10^{-3}\ \text{m}^2$$

Magnetic moment for each turn of coil is

$$\mu = IA = 5.00(7.85\times10^{-3})$$

$$= 3.93\times10^{-2}\ \text{A m}^2$$

$$\mu_{\text{total}} = 30(3.93\times10^{-2})$$

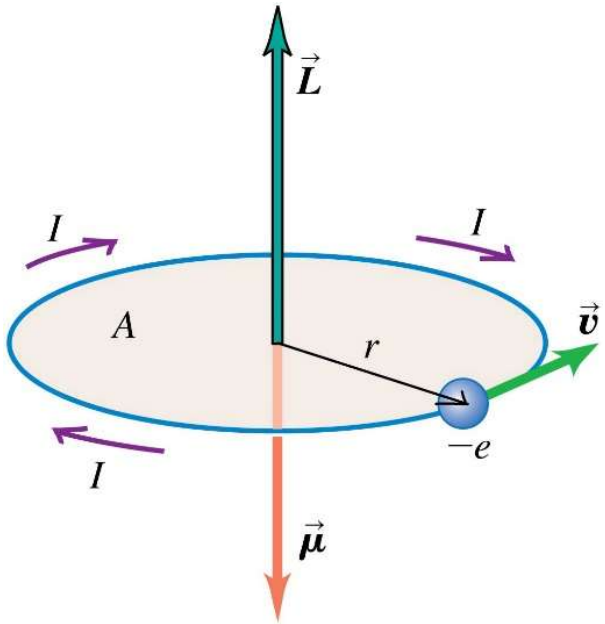
$$= 1.18\ \text{A m}^2$$

The angle ϕ between $\vec{B}$ and $\vec{\mu}$ is 90°.

$$\tau = \mu_{\text{total}} B\sin\phi = 1.41\ \text{N m}$$

The Bohr Magneton (1 of 2)

- An electron moving with speed v in a circular orbit of radius r has an angular momentum $\vec{L}$ and an oppositely directed orbital magnetic dipole moment $\vec{\mu}$.
- It also has a spin angular momentum and an oppositely directed spin magnetic dipole moment.



The Bohr Magneton (2 of 2)

- As an electron orbits the nucleus of an atom, its magnetic dipole moment has a magnitude proportional to its orbital angular momentum.

$$\mu = \frac{e}{2m} L$$

- Atomic angular momentum is **quantized**; its component in a particular direction is always an integer multiple of $\frac{h}{2\pi}$,

where $h = 6.626 \times 10^{-34}\ \text{J} \cdot \text{s}$ is Planck's constant.

- This means there is a fundamental unit of magnetic dipole moment, which is called the **Bohr magneton**,

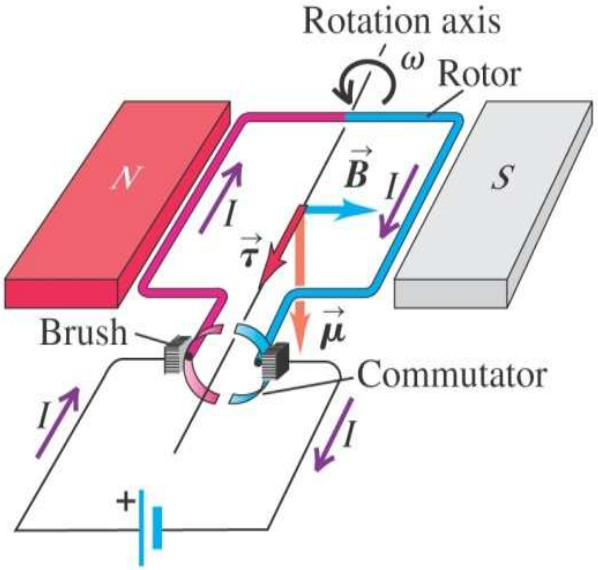
$$\mu_B = eh/4\pi m = 9.274 \times 10^{-24}\ \text{J/T}.$$

- Electron spin also leads to a magnetic moment, which is about equal to one Bohr magneton.

The Direct-Current motor

- The rotor is an open-ended loop of wire and free to rotate about an axis.
- The ends of the wire are attached to elements that form a commutator.
- Each commutator segment makes contact with one of the brushes of an external circuit that includes a source of emf.

(a) Brushes are aligned with commutator segments.



- Current flows into the red side of the rotor and out of the blue side.
- Therefore the magnetic torque causes the rotor to spin counterclockwise.

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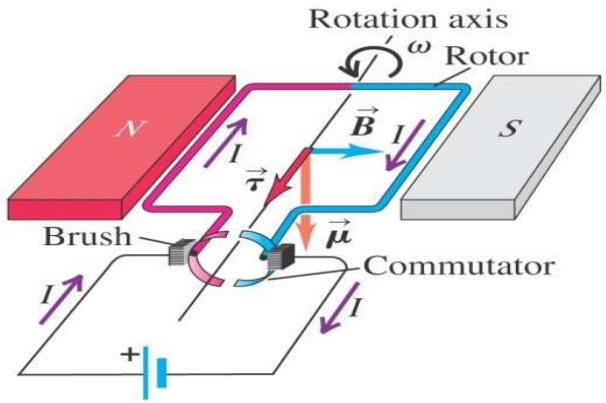
The Direct-Current motor

- This causes a current to flow into the rotor on one side and out of the rotor on the other side.

- Hence the rotor is a current loop with a magnetic moment.

- The magnetic field exerts a torque on the rotor and causes the rotor to rotate counter clockwise so that **B** is aligned with **μ**.

(a) Brushes are aligned with commutator segments.



- Current flows into the red side of the rotor and out of the blue side.
- Therefore the magnetic torque causes the rotor to spin counterclockwise.

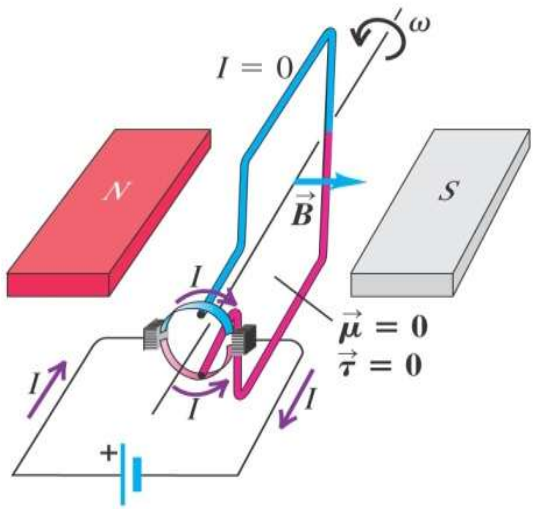
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The Direct-Current motor

- There is no potential difference between the segments of the commutator. At this instant no current flows through the rotor and the magnetic moment is zero.

- The rotor continues to rotate counter clockwise because of its inertia until current flows again through the rotor.

(b) Rotor has turned 90°.



- Each brush is in contact with both commutator segments, so the current bypasses the rotor altogether.
- No magnetic torque acts on the rotor.

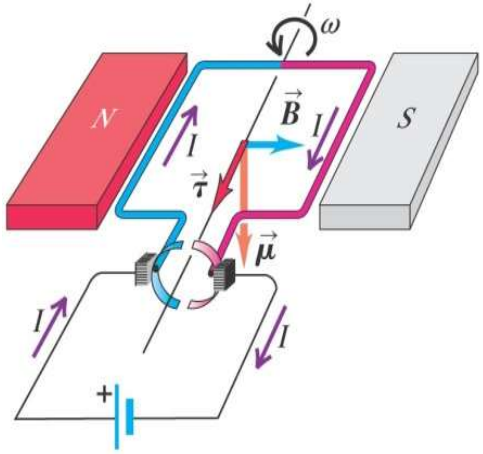
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The Direct-Current motor

- The rotor itself has rotated 180° and $\boldsymbol{\mu}$ is in the same direction with respect to **B**. The torque is in the same direction as before.

- Due to the commutator, the current reverses after every 180° of rotation so the torque is always in the direction to rotate the rotor counter clockwise.

(c) Rotor has turned 180°.

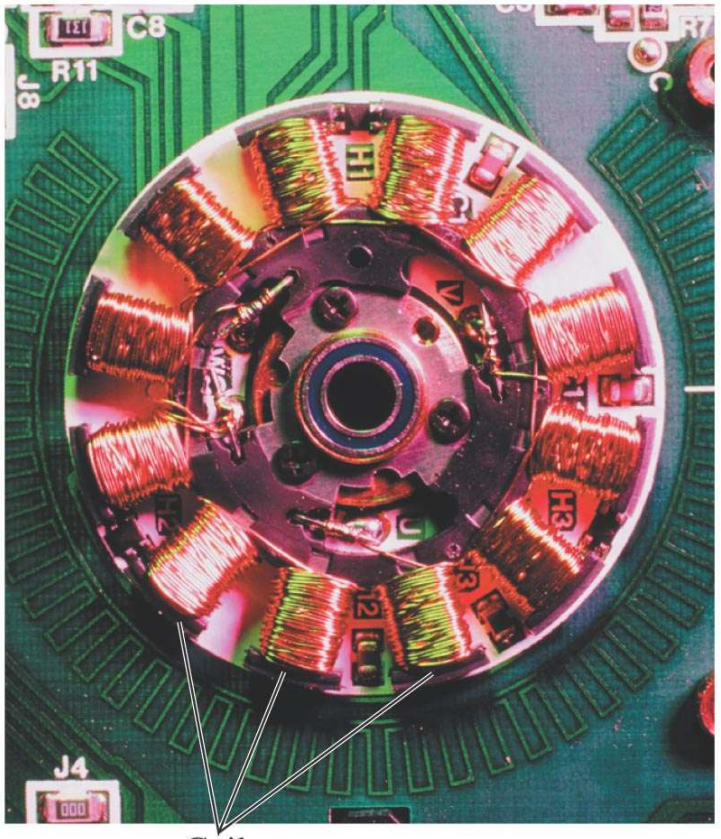


- The brushes are again aligned with commutator segments. This time the current flows into the blue side of the rotor and out of the red side.
- Therefore the magnetic torque again causes the rotor to spin counterclockwise.

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Practical motors

- In practical motors, the rotor has many turns.
- This increases the magnetic moment and the torque so that the motor can spin larger loads.
- The torque can also be increased by using a stronger magnetic field.
- Therefore many motors use electromagnets.

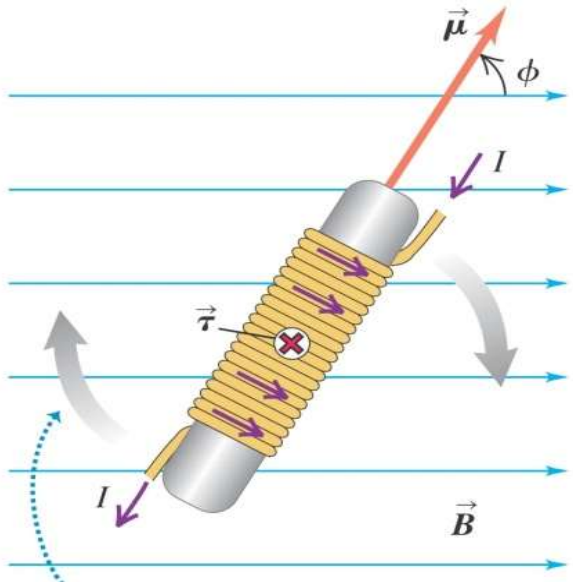


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Solenoid

- A solenoid consists of a number of circular loops lying in planes at right angles to its axis.

- For a solenoid with N turns, the magnetic moment is $\mu = NiA$. The torque is $\tau = iBA\sin\phi$ and it rotates the solenoid until its axis is parallel to the field.

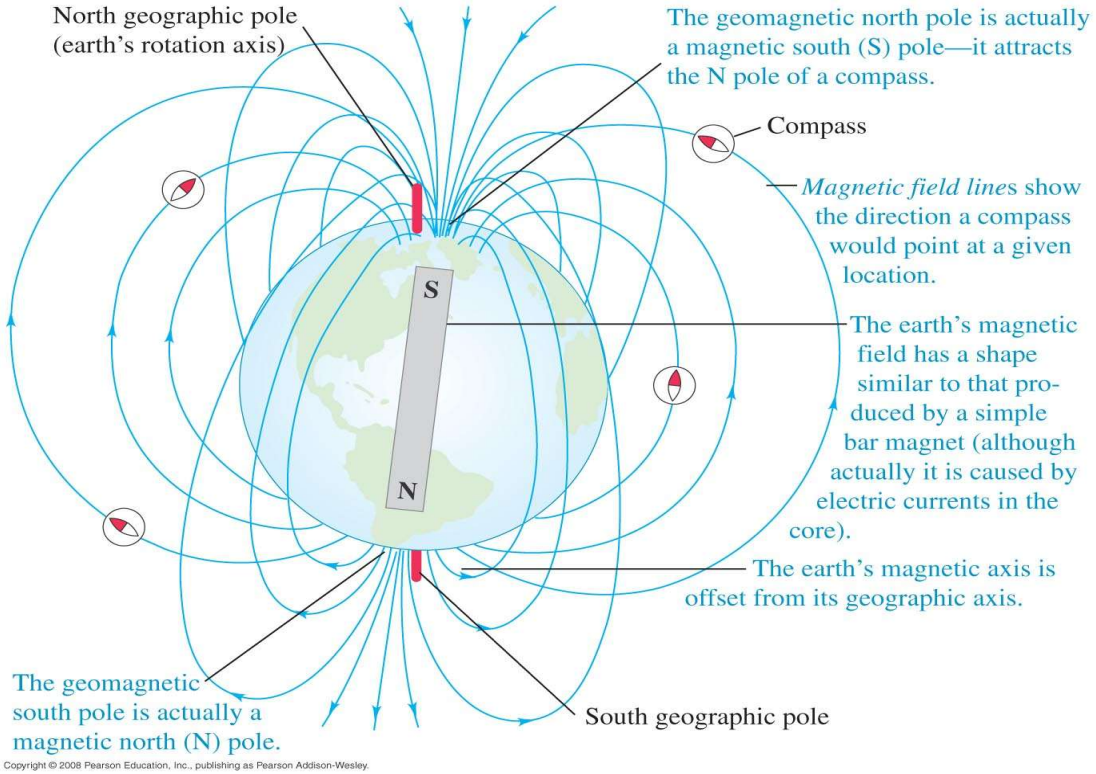


The torque tends to make the solenoid rotate clockwise in the plane of the page, aligning magnetic moment $\vec{\mu}$ with field $\vec{B}$.

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The Earth as a magnet

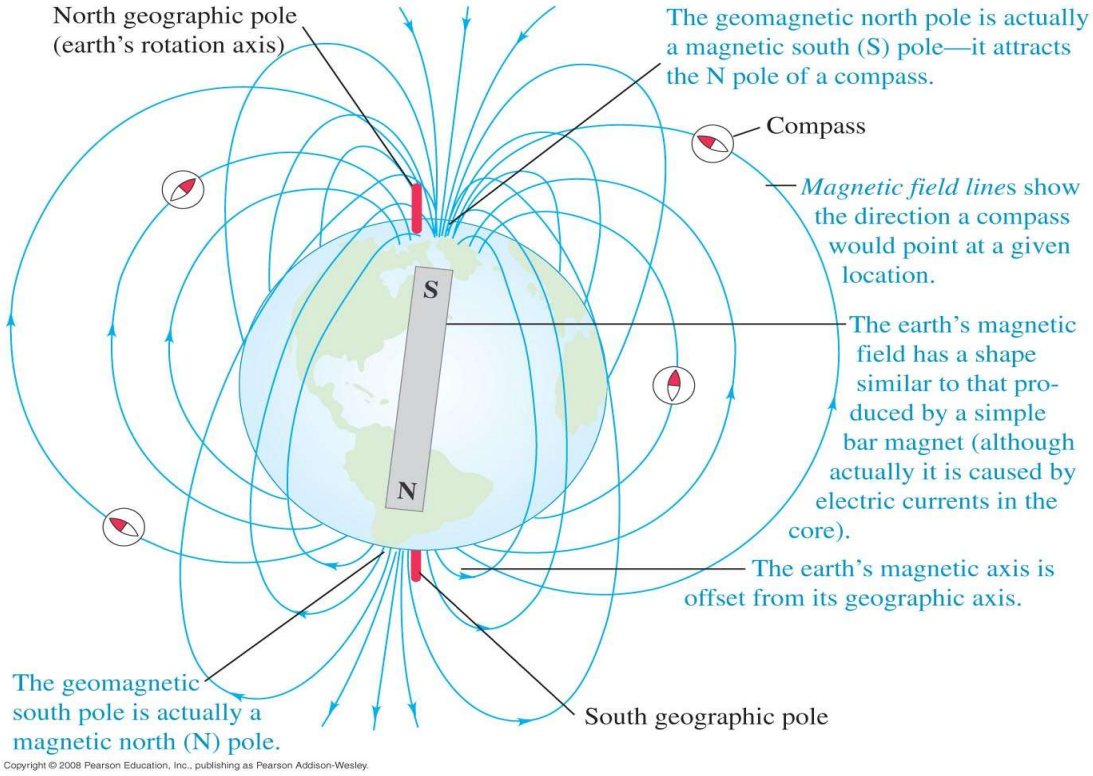
- The Earth is like a magnet with its S pole (North geomagnetic pole) in the north and its N pole (South geomagnetic pole) in the south.



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The Earth's magnetic axis

- The Earth magnetic axis is not exactly parallel to its geographic axis. Its geomagnetic poles are not at the same place as its geographic poles.
- This leads to errors known as magnetic variation.

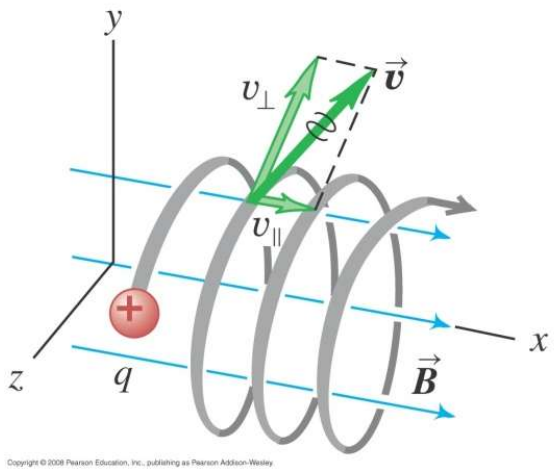


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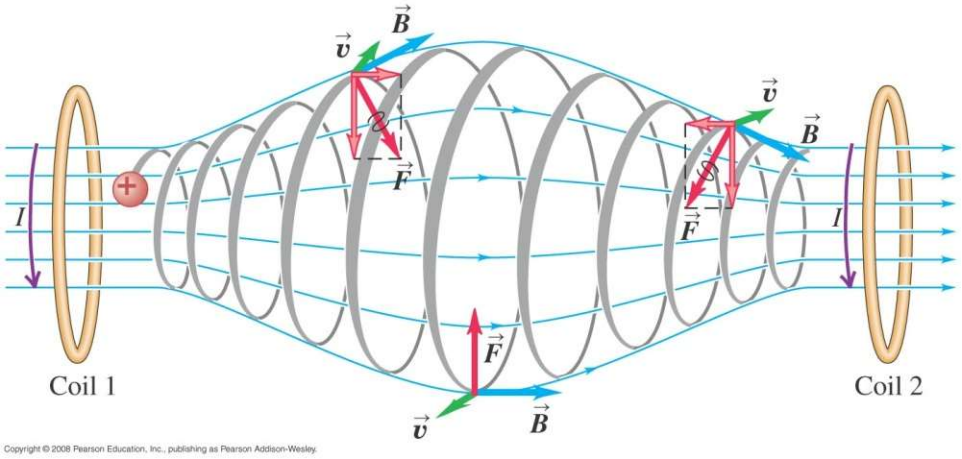
Charge particle not perpendicular to field

- If the particle's initial velocity is not perpendicular to the field, it will move in a helix with the radius we derived earlier.

- If the field is not uniform, the particles will move to and fro within the field.



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End of chapter